

Appendix C1 - Ningaloo Marine Park Data Analysis Report

North-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A1.

Benthic ecosystem extent

Both benthic habitat and demersal fish metrics are modelled with survey years pooled - the model selection results and response curves below therefore each cover a single, combined dataset rather than one figure per year.

Table C1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Survey years are pooled for this park, so covariate relationships are fitted once across all of them.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Sand	detrended+roughness+as- pect+Z	0	0.998	0.09341	8.12
Sessile invertebrates	detrended+roughness+as- pect+Z	0	0.997	0.10002	7.86
Reef	detrended+roughness+as- pect+Z	0	0.998	0.09341	8.12

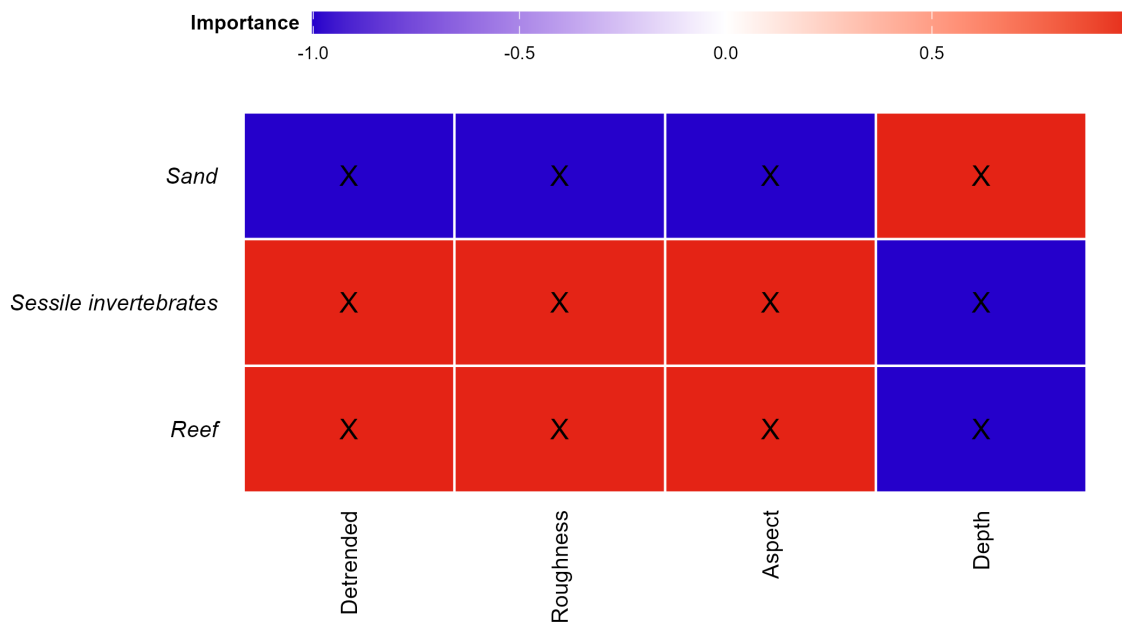


Figure C1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

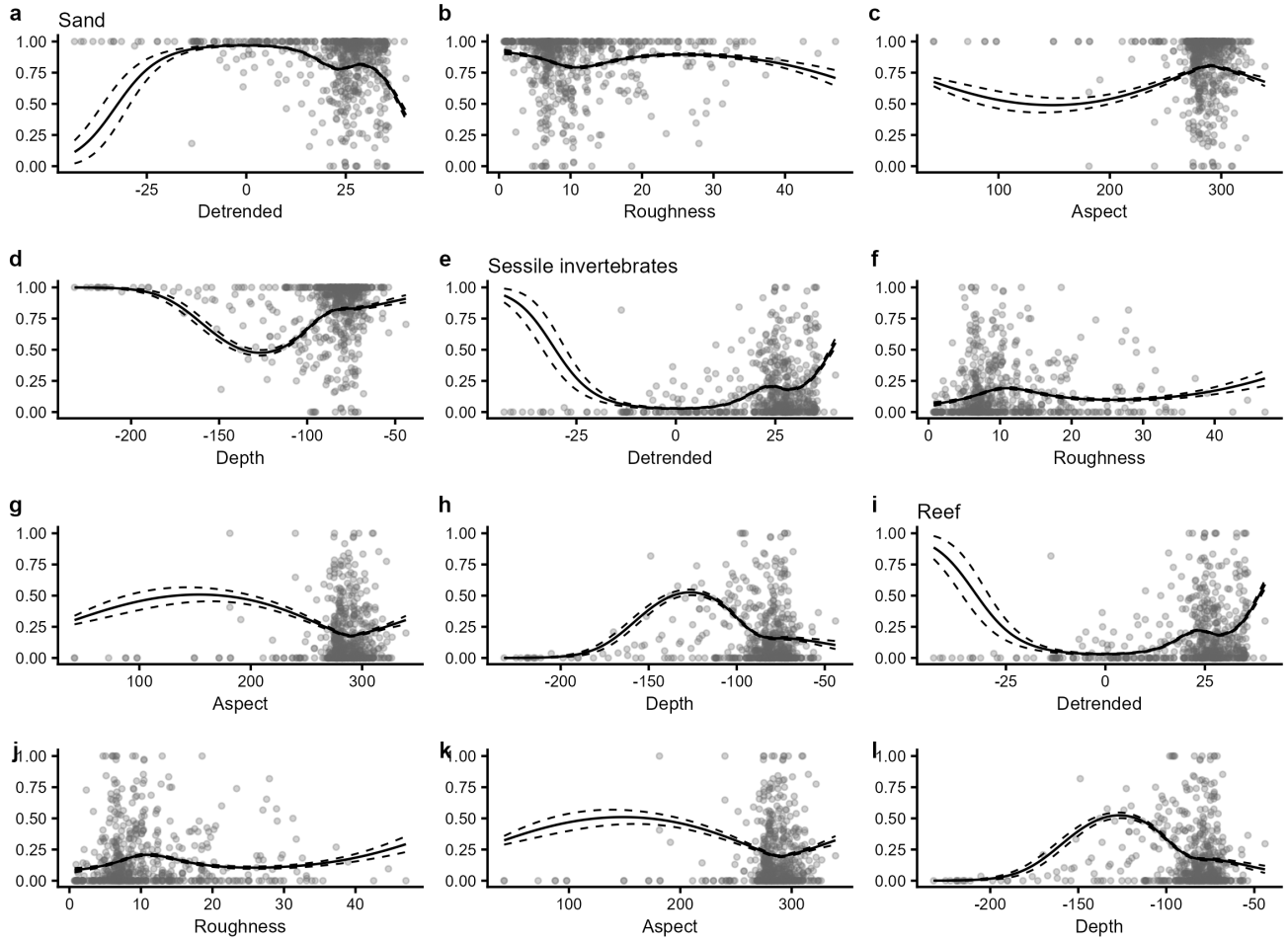


Figure C1.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data pooled across all survey years.

Demersal fish assemblage

Table C1.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. Zone status (No-Take/Fished) was forced into every candidate model rather than offered as a candidate itself. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	ΔAICc	ωAICc	AICc	R^2	EDF
Species richness	aspect+Z+reef+status	0	0.991	1780.66	0.31176	5.78
Total abundance	Z+reef+status	0	0.563	2784.17	0.10746	5.79
	aspect+Z+reef+status	0.594	0.419	2784.76	0.10989	6.79
Large Reef Fish Index*	aspect+Z+reef+status	0	0.646	6490.82	0.23148	7.25
	aspect+Z+status	1.682	0.278	6492.50	0.21857	5.69
	aspect+Z+status	0	0.275	981.46	0.15731	4.24
Reef Fish Thermal Index	aspect+Z+reef+status	0.09	0.262	981.55	0.1678	5.71
	Z+reef+status	0.101	0.261	981.57	0.1583	4.67
	Z+status	0.612	0.202	982.08	0.14497	3.00

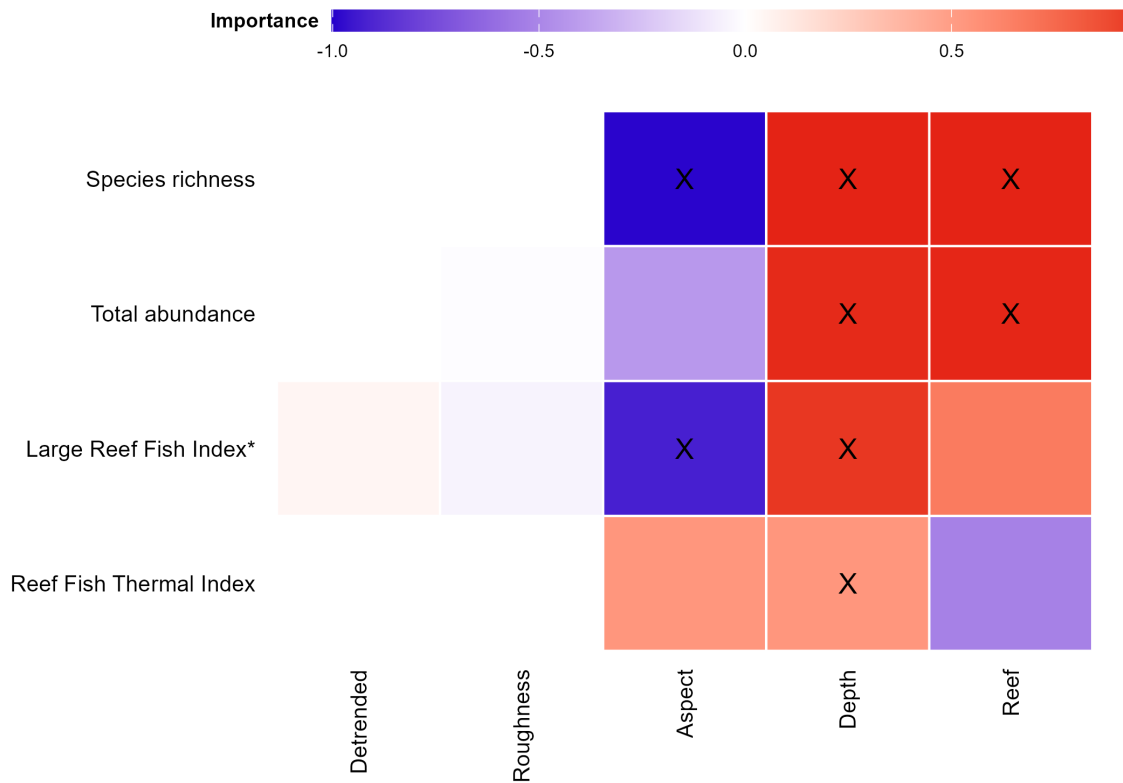


Figure C1.3: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

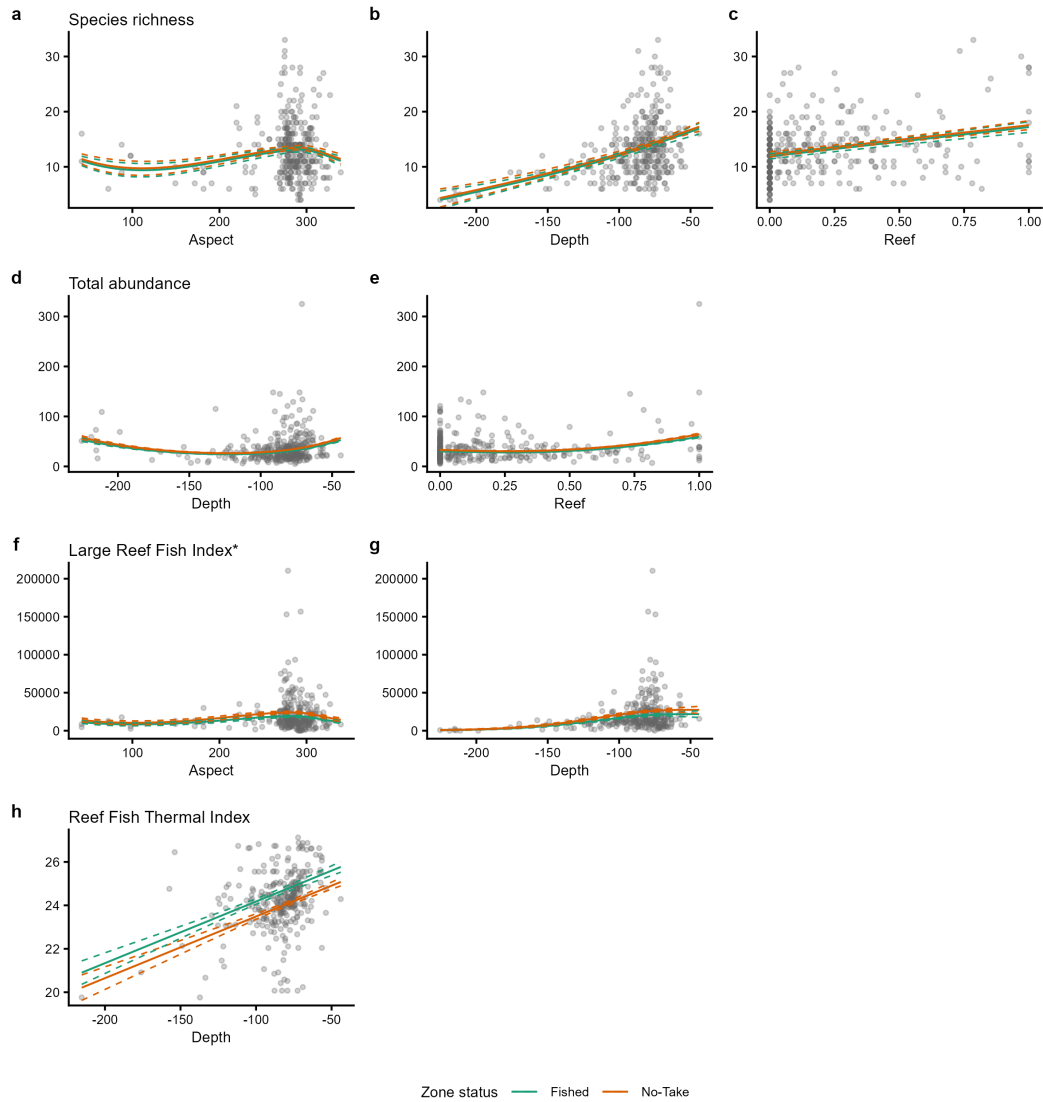


Figure C1.4: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value; line colour represents zone status (No-Take vs Fished) where status is a retained model term. Dashed lines represent upper and lower standard error bounds around each prediction, and points represent the observed data pooled across all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of zone status

Table C1.3: Significance of the zone status term in each demersal fish GAMM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary, one row per factor-level contrast (e.g. No-Take against Fished).

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Zone status: No-Take vs Fished	0.350	0.554	0.632	0.528
Total abundance	Zone status: No-Take vs Fished	0.081	0.021	3.882	<0.001
Large Reef Fish Index*	Zone status: No-Take vs Fished	0.229	0.098	2.349	0.019
Reef Fish Thermal Index	Zone status: No-Take vs Fished	-0.691	0.170	-4.065	<0.001